

Qatar-1b

R-2525

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_240917 · created 2026-08-25T01:35:59+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460570.8628 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.057
Transit depth (Rp/Rs)^2	3.63 +/- 2.17 [%]
Orbital Inclination (inc)	88.47 +/- 2.86
Airmass coefficient 1 (a1)	0.097 +/- 0.04
Airmass coefficient 2 (a2)	0.92 +/- 0.51
Scatter in the residuals of the lightcurve fit is	42.46 %
Best Comparison Star	#1 - [375, 43]
Optimal Aperture	1.32
Optimal Annulus	7.46
Transit Duration (day)	0.0825 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.057 vs 0.1463 (deviation 0.78σ)
Duration fit vs published	0.0825 d vs 0.0692 d (deviation 1.83σ)
Mid-time O–C	7.2 min
Depth SNR	3.4
Residual scatter	42.46 %

Light curve

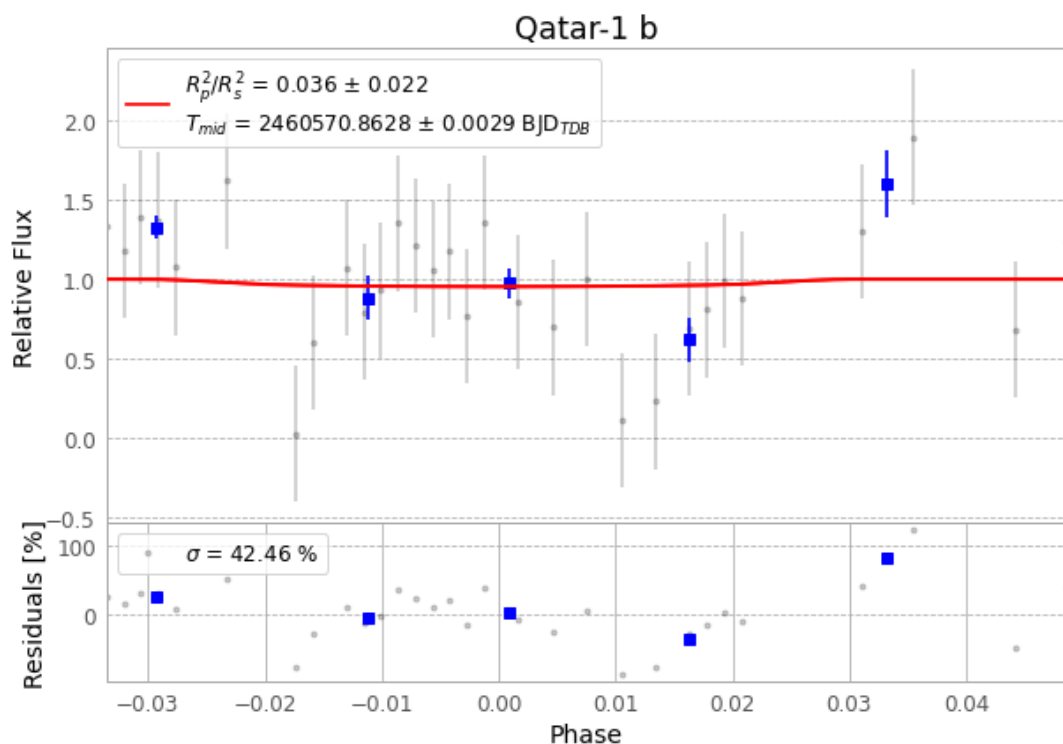


Figure 1 — FinalLightCurve_Qatar-1 b_2024-09-17.png (SHA-256 3223e257b2724e70...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [346, 296], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 107f3884becb2f74...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

59 FITS frames (39 MB), Qatar-1240917073015.FITS → Qatar-1240917102415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-240917.log (cbdace3301f4...), FinalLightCurve_Qatar-1 b_2024-09-17.png (3223e257b272...), FinalLightCurve_Qatar-1 b_2024-09-17.csv (43807d48914a...)

Compute resources

Wall time 199.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 01:36Z.